

OUR PROJECT

① PROPAGATION OF FREE GAUSSIAN WAVEPACKETS

$$\psi(0, x) = \left(\frac{2\alpha}{\pi}\right)^{\frac{1}{4}} e^{-\alpha x^2} e^{+\frac{i}{\hbar} p x}$$

↓ FT

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} dx \psi(0, x) e^{-\frac{i}{\hbar} p x} = \left(\frac{1}{\pi\hbar^2 2\alpha}\right)^{\frac{1}{4}} e^{-\frac{1}{4\hbar^2 \alpha} (p - p)^2}$$

$$\psi(t, x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} dp \phi(p) e^{+\frac{i}{\hbar} p x} e^{-\frac{i}{\hbar} \frac{p^2}{2m} t}$$

② REGULARIZED DIRAC δ -FUNCTION

$$\frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{+i\frac{x}{L}} d\frac{x}{L} = \delta\left(\frac{x}{L}\right)$$

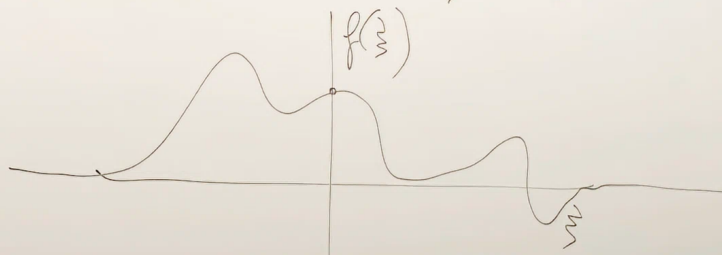
↓ REGULARIZATION

$$\frac{1}{2\pi} \int_{-L}^{+L} e^{+i\frac{x}{L}} d\frac{x}{L} = \delta_L\left(\frac{x}{L}\right)$$

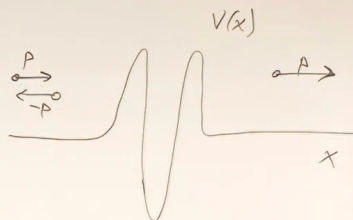
$$L \rightarrow +\infty$$

$$\lim_{L \rightarrow \infty} \int_{-\infty}^{+\infty} d\frac{x}{L} f\left(\frac{x}{L}\right) \delta_L\left(\frac{x}{L}\right) = f(0)$$

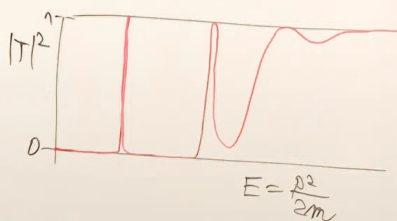
FOR ANY C^0 TEST FUNCTION
VANISHING AT $\frac{x}{L} \rightarrow \pm\infty$



③ TRANSMISSION/ REFLECTION PHENOMENA



$$E = \frac{p^2}{2m}$$



$$\hat{H} = -\frac{\hbar^2}{2m} \partial_{xx} + V(x)$$

$$\hat{H} \psi_p(x) = E \psi_p(x)$$

$$E = \frac{p^2}{2m}$$

$$\hat{H} \psi_0(x) = E_0 \psi_0(x)$$

$$\hat{H} \psi_1(x) = E_1 \psi_1(x)$$

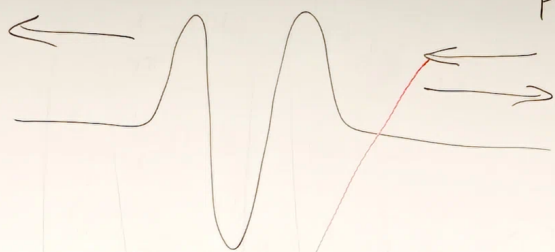
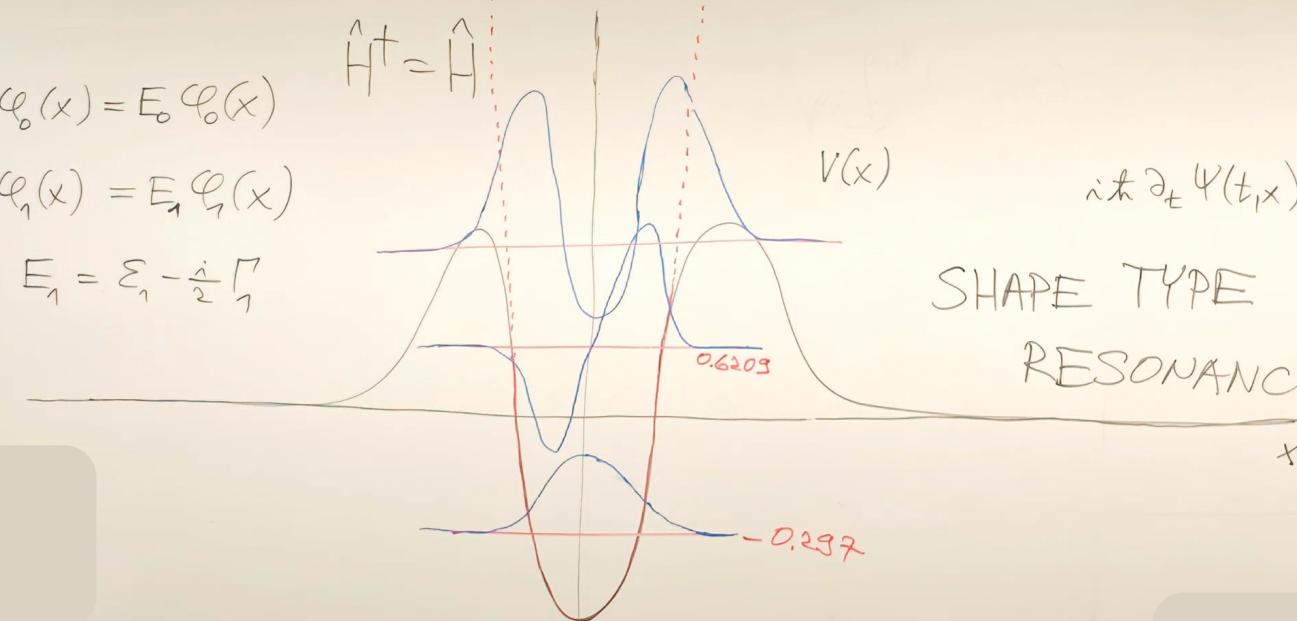
$$E_1 = E_0 - \frac{i}{2} \Gamma$$

$$\hat{H}^\dagger = \hat{H}$$

$V(x)$

$$i\hbar \partial_t \psi(t,x) = \hat{H} \psi(t,x)$$

SHAPE TYPE
RESONANCES



$$\frac{1}{\sqrt{2\pi\hbar}} e^{+i\frac{p}{\hbar}x}$$

$p < 0$

$$\hat{H} \psi_p(x) = E \psi_p(x)$$

$E > 0$

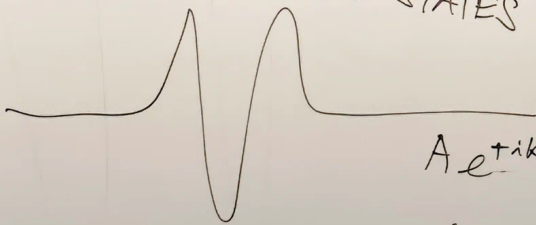
$$-\frac{\hbar^2}{2m} \partial_{xx} \psi_p(x) + V(x) \psi_p(x) = E \psi_p(x) \quad p > 0$$

BOUNDARY CONDITIONS:

$$\psi_p(x \rightarrow +\infty) = T(p) \frac{1}{\sqrt{2\pi\hbar}} e^{+i\frac{p}{\hbar}x}$$

$$\psi_p(x \rightarrow -\infty) = \frac{1}{\sqrt{2\pi\hbar}} e^{+i\frac{p}{\hbar}x} + R(p) \frac{1}{\sqrt{2\pi\hbar}} e^{-i\frac{p}{\hbar}x}$$

WE HAVE NOW IN HAND: $\psi_p(x)$
CONTINUUM STATES $\psi_p(x)$
SCATTERING STATES $\psi_p(x)$



$$A e^{+ikx} + B e^{-ikx}$$

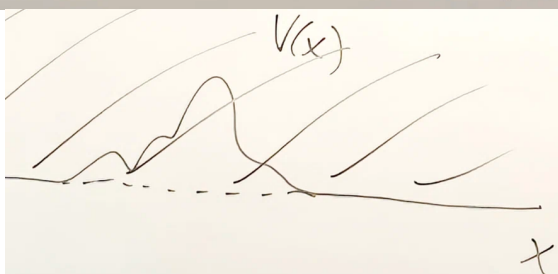
$$\hbar k = \sqrt{2mE} > 0$$

$p > 0$

$p < 0$

FOR ANY $E > 0$

WE HAVE TWO DEGENERATE
SOLUTIONS OF OUR TISE
NOT SQUARE INTEGRABLE



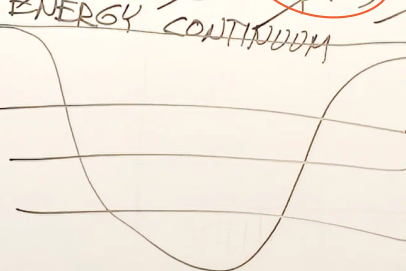
$E > 0 \dots$

DOUBLY
DEGENERATE
ENERGY CONTINUUM

$\psi_p(x)$

$$p = \pm \sqrt{2mE}$$

$\psi_j(x)$



$$\hat{H}\psi(x) = E\psi(x)$$

BOUND STATES $\psi_j(x)$, $E_j < 0$

ALWAYS A FINITE NUMBER
(THERE CAN ALSO BE NONE)

CONTINUUM STATES $\psi_p(x)$ $p > 0$
or $p < 0$

$$E = \frac{p^2}{2m} > 0$$

ORTHONORMALITY

$$\int_{-\infty}^{+\infty} dx \psi_j^*(x) \psi_{j'}(x) = \delta_{jj'}$$

$$\int_{-\infty}^{+\infty} dx \psi_p^*(x) \psi_{p'}(x) = \delta(p-p')$$

$$\int_{-\infty}^{+\infty} dx \psi_j^*(x) \psi_p(x) = 0$$

$$\int_{-\infty}^{+\infty} dx \psi_p^*(x) \psi_{j'}(x) = 0$$

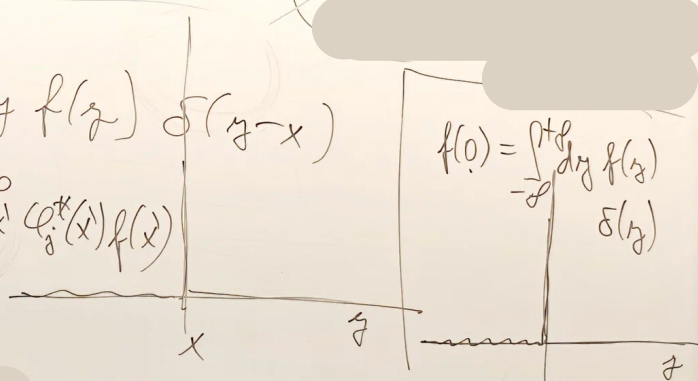
CLOSURE

$$\sum_j \psi_j(x) \psi_j^*(x')$$

$$+ \int_{-\infty}^{+\infty} dp \psi_p(x) \psi_p^*(x') = \delta(x-x')$$

$$\begin{aligned} f(x) &= \int_{-\infty}^{+\infty} dx' f(x') \delta(x-x') = \\ &= \int_{-\infty}^{+\infty} dx' f(x') \sum_j \psi_j(x) \psi_j^*(x') = \\ &= \sum_j c_j \psi_j(x) \end{aligned}$$

$$\begin{aligned} f(x) &= \int_{-\infty}^{+\infty} dy f(y) \delta(y-x) \\ c_j &= \int_{-\infty}^{+\infty} dx' \psi_j^*(x') f(x') \end{aligned}$$



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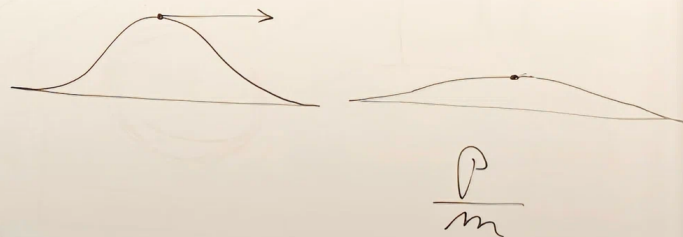
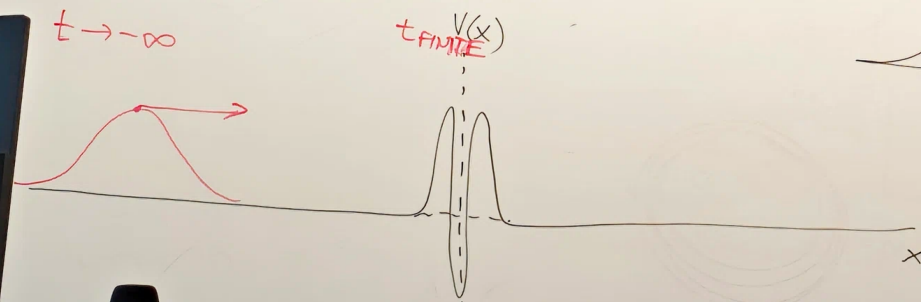
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↓ FT

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$$i\hbar \partial_t \psi(t, x) = \frac{\hat{p}^2}{2m} \psi(t, x)$$



PART 4 OF OUR PROJECT:

$$i\hbar \partial_t \psi(t, x) = \left(\frac{\hat{p}^2}{2m} + V(x)\right) \psi(t, x)$$

$$\psi(0, x) = \psi_0(x) \text{ GIVEN}$$

$$\psi_0(x) = \int_{-\infty}^{+\infty} dx' \psi_0(x') \delta(x - x') =$$

$$= \int_{-\infty}^{+\infty} dx' \psi_0(x') \sum_j c_j \varphi_j(x) \varphi_j^*(x')$$

$$+ \int_{-\infty}^{+\infty} dx' \psi_0(x') \int_{-\infty}^{+\infty} dp \varphi_p(x) \varphi_p^*(x') =$$

$$= \sum_j c_j \varphi_j(x)$$

$$+ \int_{-\infty}^{+\infty} dp \phi(p) \varphi_p(x)$$

$$c_j = \int_{-\infty}^{+\infty} dx \varphi_j^*(x) \psi_0(x)$$

$$\phi(p) = \int_{-\infty}^{+\infty} dx \varphi_p^*(x) \psi_0(x)$$

DIRECT CALCULATION:

$$\langle \phi | \phi \rangle = 1$$

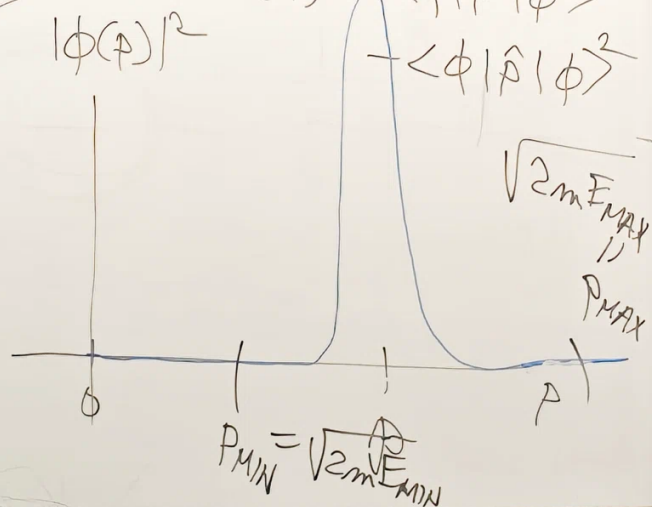
$$\langle \phi | \hat{p} | \phi \rangle = p$$

UNCERTAINTY

$$\Delta p = \hbar \sqrt{\alpha}$$

$$(\Delta p)^2 = \langle \phi | \hat{p}^2 | \phi \rangle$$

$$- \langle \phi | \hat{p} | \phi \rangle^2$$



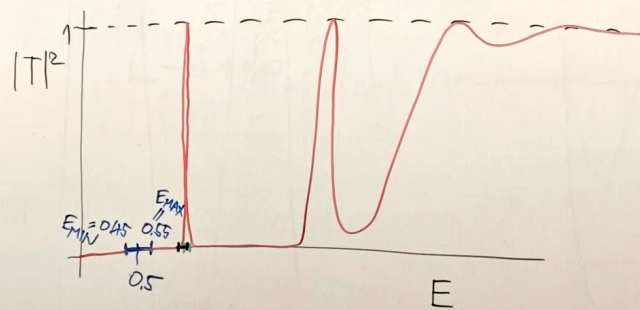
PART 4 OF OUR PROJECT:

$$i\hbar \partial_t \psi(t,x) = \left(\frac{\hat{p}^2}{2m} + V(x) \right) \psi(t,x)$$

$$\psi(0,x) = \psi_0(x) \text{ GIVEN}$$

FOR OUR CASE:

WAVEPACKET PICTURE OF A QUANTUM COLLISION (SCATTERING)



$$E = 0.5$$

$$\frac{p^2}{2m} = 0.5$$

$$m=1$$

$$\psi(t,x) = \sum_j c_j \phi_j(x) e^{-\frac{i}{\hbar} E_j t} + \int_{-\infty}^{+\infty} dp \phi(p) \psi_p(x) e^{-\frac{i}{\hbar} \frac{p^2}{2m} t}$$

$c_j = 0 \quad \forall j$

